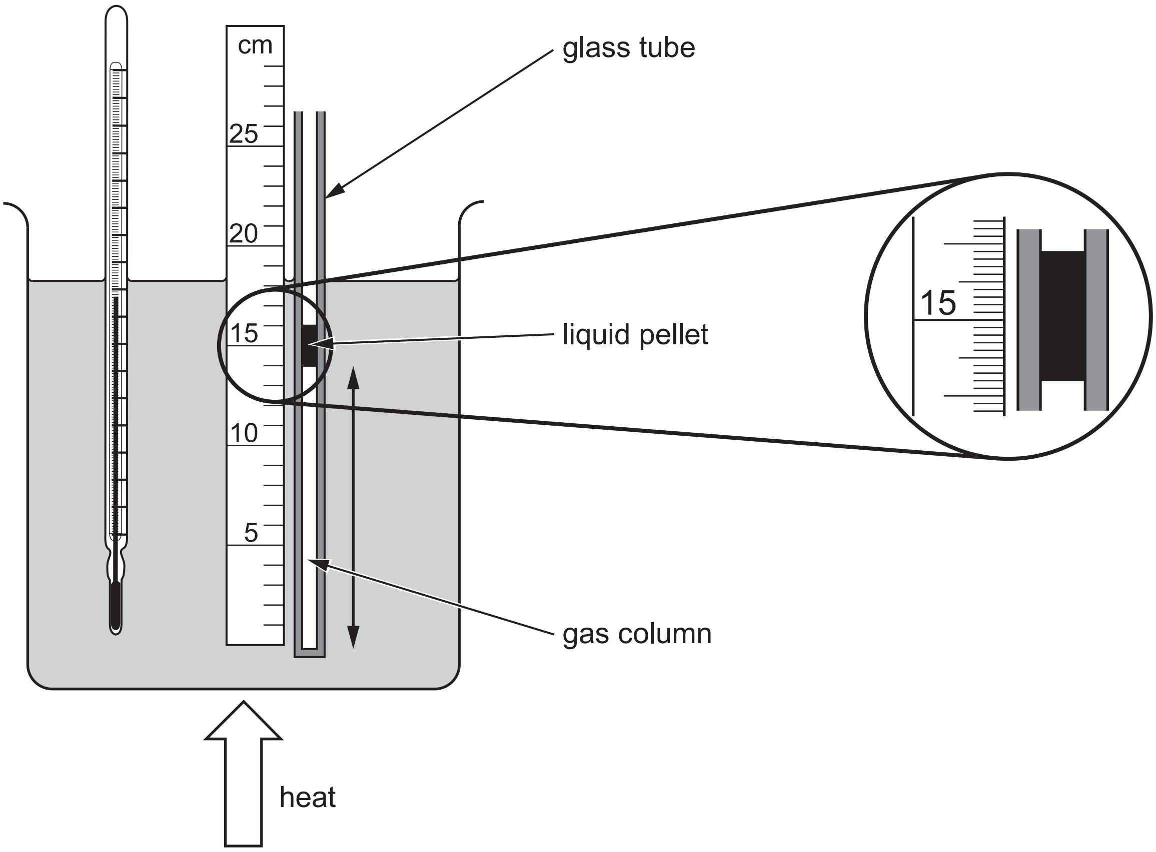
SECTION A

Answer **all** *questions.*

1. A student performs an experiment to estimate the Celsius temperature of absolute zero.



The length, l, of the gas column and its temperature, θ, are measured at atmospheric pressure, 1.01×10^5 Pa. The temperature of the water is initially 0.0 °C. It is then increased to 80.0 °C. Readings are taken at 20.0 °C intervals. The scale used to measure the length gives an uncertainty of ± 0.1 cm at the top of the column and an uncertainty of ± 0.1 cm at the bottom.

Values of θ and l are recorded in the table below.

θ / °C	l / cm
0.0	11.5
20.0	12.5
40.0	13.2
60.0	14.2
80.0	15.0

Examiner only

(a) Justify the number of significant figures used to record the length, l. [2]

....

....

....

(b) The glass tube in which the gas is trapped has an internal diameter of (1.5 ± 0.1) mm.

(i) Calculate the volume of the gas at 0.0 °C. [2]

....

....

....

....

....

....

(ii) Show that the percentage uncertainty in this volume, V, is approximately 15%. [3]

....

....

....

....

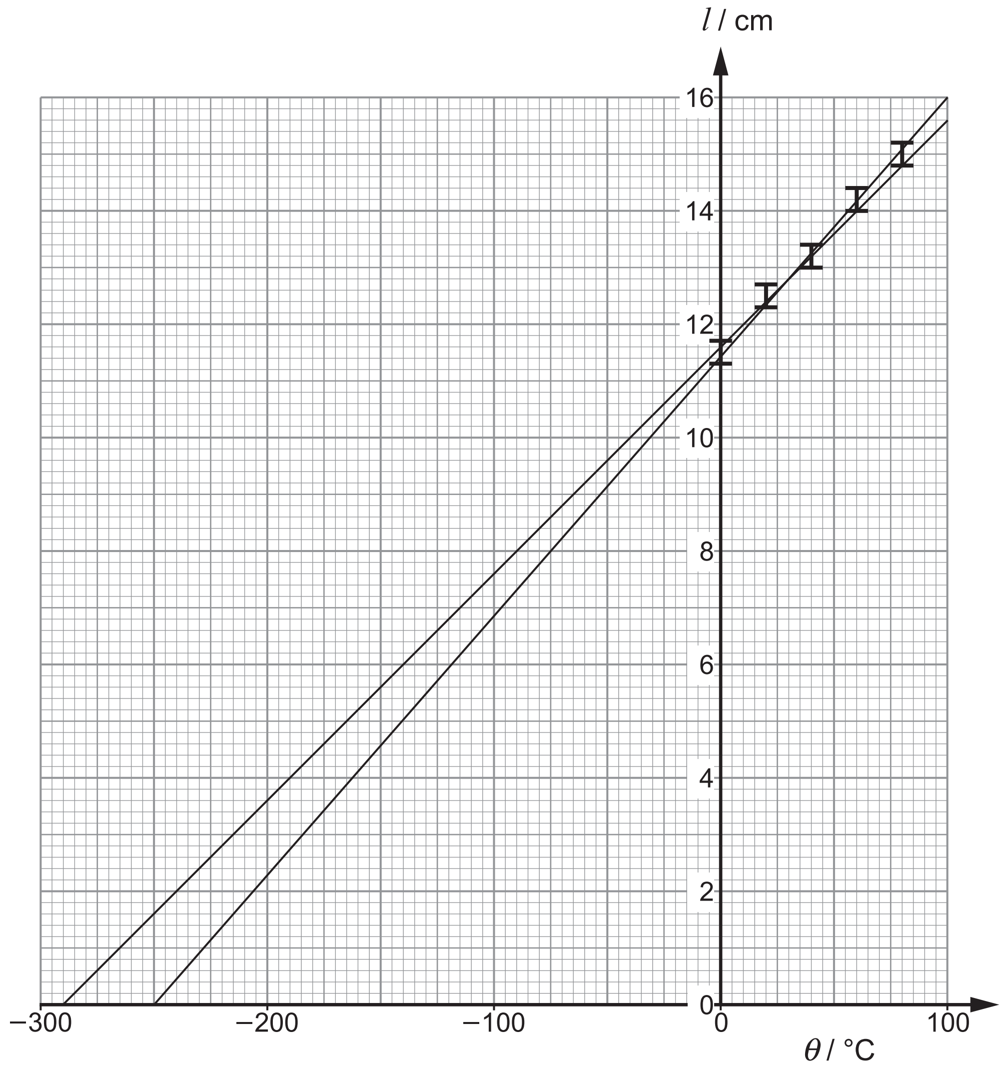
....

....

1420U301 03

Examiner only

(c) The length, l, is plotted against the temperature, θ. Lines of maximum and minimum gradient consistent with the error bars have been drawn.



(i) Use the graph to estimate the temperature of absolute zero in °C, together with its **absolute** uncertainty. [2]

....

....

....

Examiner only

(ii) Justify the statement that *the results are consistent with the ideal gas equation.* [4]

....

....

....

....

....

....

....

....

....

(d) State what happens to the behaviour of the gas molecules as the temperature approaches absolute zero. [1]

....

....

....

(e) The scale resolution of the ruler affects the final uncertainty in the value of absolute zero. Identify **two** other factors that could affect the accuracy. [2]

....

....

....

1420U301 05

16